

- ERC-20-wrapped version of ETH to get around this. Uniswap allows a user to directly supply and trade with ETH and converts to WETH behind the scenes. See “WTF Is WETH?” Radar Relay, Inc., 2021, <https://weth.io/>.
16. <https://github.com/bogatyy/bancor>
 17. <https://explore.flashbots.net/>
 18. This is a smart contract level check. In other words, before finalizing the trade, the contract checks the total slippage from the initially posted price to the effective execution price (which could have changed if other transactions made it in first like the described front-running attempt). If this slippage exceeds the predefined user tolerance, the entire trade is canceled, and the contract execution fails.
 19. Andrey Shevchenko, “A New DeFi Exchange Says It Has Solved an Industry-Wide Problem,” *Cointelegraph*, August 11, 2020, <https://cointelegraph.com/news/a-new-defi-exchange-says-it-has-solved-an-industry-wide-problem>.
 20. Sushiswap, <https://sushi.com/>.
 21. *Balancer*, Balancer Labs, <https://balancer.finance/>.
 22. The bonding surface in Balancer is given by $V = \prod_{t=0}^n B_t^{W^t}$, where V is the value function (analogous to k), n is the number of assets in the pool, B is the balance of the token t in the pool, and W is the normalized weight of token t . See Fernando Martinelli, “Bonding Surfaces & Balancer Protocol,” *Balancer*, October 4, 2019, <https://>